ASALI (असली) vs NAKALI (नकली)

The Complete Zero-to-100 Master Architecture, Codebase, and Theory Guide

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1. Executive Overview & The Real-World Crisis

In the pharmaceutical supply chain, counterfeit medications (**Nakali**) kill hundreds of thousands of people each year. Substandard drugs range from:

- **Chalk & Starch Fakes**: Packaging printed to look identical to Cipla Novamox, but with zero active pharmaceutical ingredient (API).
- **Toxic Contamination**: Medicines formulated with lethal industrial solvents (such as Diethylene Glycol in cough syrups).
- **Barcode Cloning / Digital Duplication**: A counterfeiter buys one legitimate box, copies its 2D DataMatrix code onto 10,000 fake boxes, and distributes them simultaneously across hundreds of pharmacies.
- **Cold-Chain Spoilage**: Vaccines, insulins, or biologicals that experienced heat excursion ($> 8^\circ\text{C}$) during transport, denaturing the protein structure and rendering the drug ineffective or toxic.
- **Black Market / Illicit Diversion**: Medicines stolen from hospital dispensaries or routed through unlicensed vendors.

Asali (Swahili/Hindi for *"Pure / Genuine / Authentic"*) is a federated Information Integration platform engineered to verify any medication pack in real time at point-of-sale or hospital dispensation.

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2. Why Not a Single Monolithic Database? (The Core IIA Problem)

Beginners often ask:

> *"Why not just put all pharmaceutical data into one giant Oracle or PostgreSQL database in the cloud?"*

In the real world, a single database is **technically, commercially, and legally impossible**:

Autonomous Enterprise Boundaries:

Manufacturers (e.g., Cipla, Sun Pharma, Pfizer) treat their batch release formulas and ERP cost data as trade secrets. They will never give direct database write access to external logistics companies.

Logistics & Freight Forwarders run their own warehouse management systems (often SAP or custom relational DBs).

Retail Dispensaries / Pharmacies run thousands of independent Point-of-Sale (POS) systems (often streaming document stores or lightweight NoSQL engines).

Government Regulatory Agencies (CDSCO, Ministry of Consumer Affairs) maintain statutory master lists and blacklists completely isolated from commercial networks.

System & Syntactic Heterogeneity:

Different stakeholders choose different databases optimized for their specific workload:

Retail POS terminals require high-velocity NoSQL document streams.

Logistics ERP requires 3NF relational databases with foreign keys.

Central regulators publish secure REST APIs.

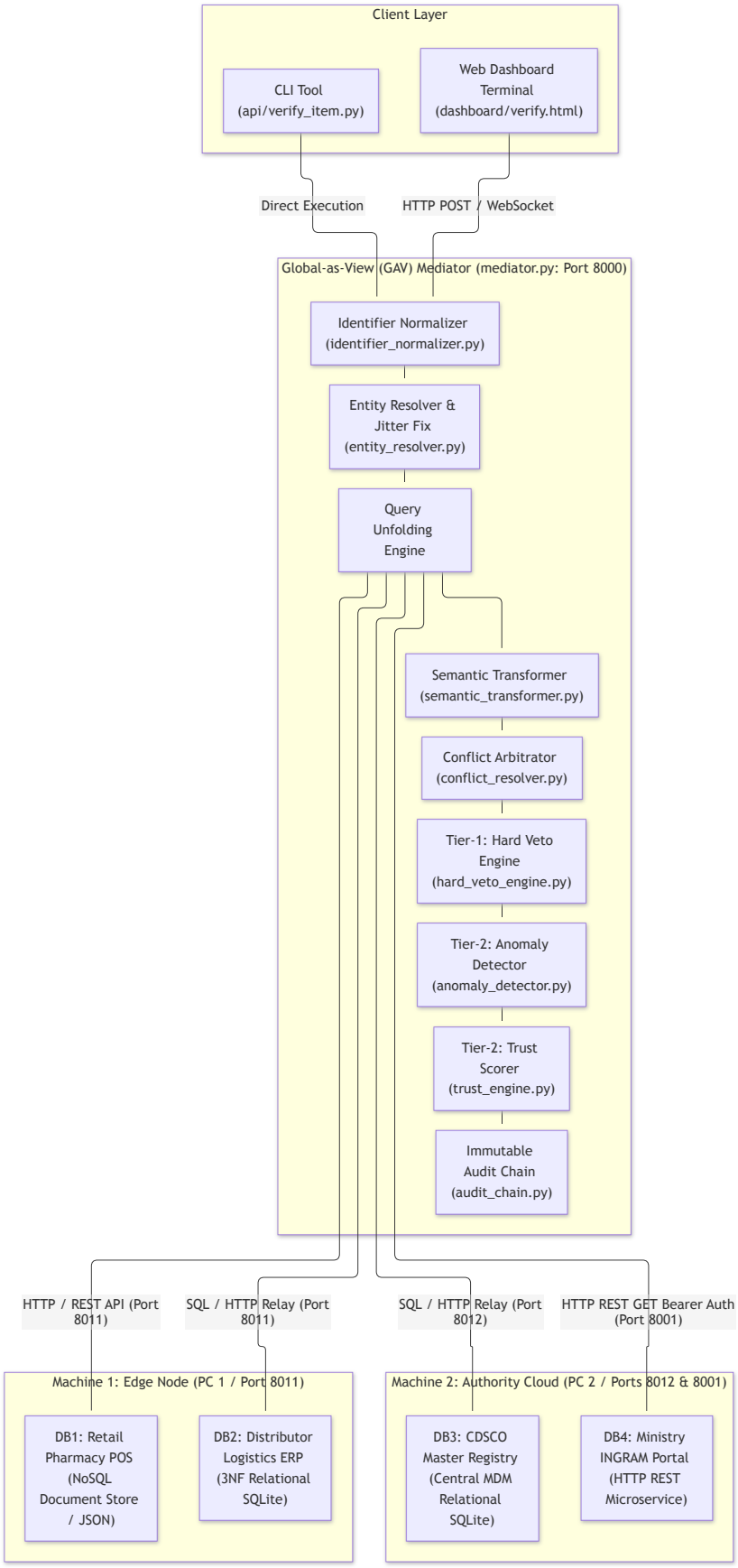
Regulatory & Geographic Sovereignty:

Data residency laws and antitrust regulations prevent centralized commercial monopolies over health data.

The Solution: **Global-as-View (GAV) Federated Query Mediation**.
Instead of centralizing data, Asali centralizes **logic**. The databases remain completely autonomous, heterogeneous, and distributed. The Asali Mediator issues distributed subqueries on the fly, translates syntaxes, resolves typos, arbitrates contradictions, and calculates a unified verdict.

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3. Global Architecture & Distributed Mesh Topology



The system operates across **3 tiers**:

Client Tier: Modern web application (`index.html` , `verify.html`) and Python automation CLI (`api/verify_item.py`).

Mediation Tier: The federated query coordinator, running the GAV resolution pipeline and the Two-Tier verification engine.

Data Witness Tier: 4 physically independent, heterogeneous database nodes.

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4. Data Generation & Seeding Pipeline (Zero-to-One)

All data in Asali is deterministically generated from scratch to model realistic pharmaceutical scenarios.



Step 1: Schema Creation (`create_databases.py`)

Reads schema definition files from `schema/` :

- `schema_db1.sql` → DB1
- `schema_db2.sql` → DB2
- `schema_db3.sql` → DB3
- `schema_db4.sql` → DB4

Deletes any stale SQLite databases and compiles clean binary database files in `database s/` .

Step 2: Relational Seeding (`data/seed_db*.py`)

`seed_db3.py` **(CDSCO Manufacturer Registry)**:

Creates 5 licensed Indian pharmaceutical giants: `MFG-CIPLA-01`, `MFG-SUN-02`, `MFG-DR REDDY-03`, `MFG-LUPIN-04`, `MFG-ZYDUS-05`.

Injects product catalog records: `Novamox 500mg`, `Pantocid 40mg`, `Ciplox 500mg`, `Azithral 500mg`, `Insugen 40IU`.

Attaches ISO/GMP certificates and batch QC release signatures (`qc_passed = 1`).

`seed_db2.py` **(Distributor Logistics ERP)**:

Creates cold-chain freight companies: `DIST-APOLLO-LOG`, `DIST-TRANSMED-EXP`, `DIST-BLUE-DART-PHARMA`.

Generates distribution batches with transport vehicle numbers, source warehouses, destination pharmacy IDs, and GPS telemetry URLs.

`seed_db1.py` **(Point-of-Sale Scans)**:

Creates licensed pharmacies (`VND-MUM-001` Apollo Mumbai, `VND-BLR-002` MedPlus Bengaluru) and unlicensed rogue kiosks (`VND-999` Unlicensed Chemist).

Injects scan records with timestamps, prices, payment methods (UPI, Cash), and device IDs.

`seed_db4.py` **(Ministry of Consumer Affairs & INGRAM)**:

Injects consumer complaints, forensic lab reports (chemical assay potency match), and the statutory `ministry_blacklist`.

Injected Test Cases:

Test Code	Scenario Type	Why It Exists & What It Proves
`MED-ASALI-001`	**Genuine Baseline**	Verified in DB3, authorized in DB2, scanned legally in DB1, clean in DB4. Score: 100/100 (**ASALI**).
`MED-ASALI-002`	**Barcode Cloning**	Scanned in 5 different stores across Mumbai, Pune, and Bengaluru in 28 minutes. Proves geospatial velocity contradiction ($> 800\text{ km/h}$).
`MED-RECALL-001`	**Toxic Contamination**	Active recall in DB3 and gazetted blacklist in DB4. Proves Authority Precedence Policy.
`MED-UNAUTH-001`	**Black Market Diversion**	Sold by unlicensed kiosk `VND-999` with zero distributor records in DB2. Proves Supply Chain Gap detection.
`MED-ASALI-009`	**Cold-Chain Excursion**	Temperature spike to 14.2°C ($> 8^\circ\text{C}$). Proves Semantic Unit Normalization and potency risk penalty.
`MED-NAKALI-001`	**Phantom Fake**	Sold at retail in DB1, but **zero manufacturer record in DB3**. Proves Tier-1 Hard Veto.

Step 3: Native NoSQL Document Conversion (`data/seed_nosql_db1.py`)

In modern retail, thousands of POS cash registers stream semi-structured JSON directly into document stores like MongoDB.
`seed_nosql_db1.py` converts DB1 relational rows into:
`data/nosql_store/db1_retail_pos.json`
with nested document trees:

```
{
  "_id": "EVT-MUM-001",
  "scan_code": "MED-ASALI-001",
  "document_type": "retail_pos_scan",
  "dispensary": {
    "vendor_id": "VND-MUM-001",
    "name": "Apollo Pharmacy Bandra",
    "geo": { "city": "Mumbai", "coordinates": { "lat": 19.0760, "lon": 72.8777 } }
  },
  "transaction": {
    "amount": 48.0,
    "currency": "INR",
    "payment": { "method": "upi", "device_id": "POS-MUM-01" }
  },
  "quality_assurance": { "anomaly_flag": 0, "verified_at_counter": true }
}
```


---

5. The 4 Heterogeneous Databases: Schemas & Meanings

Database 1: Retail Purchase Registry (DB1)

Role: High-velocity retail scan events at point of sale.

Paradigm: NoSQL Document Store (JSON) with relational SQLite backup.

Primary Integration Key: `scan_code`.

Core Entities:

`vendors`: Store ID, name, license number, license expiry date, active status.

`purchase_events`: Scan code, scan timestamp, vendor ID, price paid, payment method, device ID.

`scan_anomaly_log`: Rapid rescan alerts, multi-location flags.

Database 2: Distributor Logistics Ledger (DB2)

Role: Tracks middle-mile transportation from factory to pharmacy.

Paradigm: 3NF Relational SQL (SQLite).

Primary Integration Key: `batch_serial`.

Core Entities:

`distributors`: Freight license number, cold-chain capability flag, authorization status.

`distribution_batches`: Batch serial, distributor ID, source warehouse, destination vendor ID, dispatch date, arrival date, `cold_chain_maintained` flag, temperature log URL, vehicle registration.

`distributor_vendor_contracts`: Legal contracts between logistics firms and pharmacies.

`return_log`: Damaged or rejected batch shipments.

Database 3: CDSCO Master Manufacturer Registry (DB3)

Role: The statutory Golden Record of genuine medicines.

Paradigm: Central Master Data Management (MDM) Relational Catalog (SQLite).

Primary Integration Key: `product_uid`.

Core Entities:

`manufacturers`: Manufacturer ID, company name, GMP certification number, license expiry.

`products`: Product UID, brand name, generic formulation, strength, batch number, manufacture date, expiry date, registration number, `requires_cold_chain` flag, `is_recalled` flag, MSRP.

`authorized_distributors`: Whitelist of approved logistics partners per manufacturer.

`batch_release_certs`: QC laboratory release certificate with cryptographic hash.

Database 4: Ministry of Consumer Affairs Grievance Portal (DB4)

Role: Government oversight, consumer complaints, and legal blacklists.

Paradigm: HTTP RESTful Web Microservice (`api/ministry_service.py` on Port 8001).

Primary Integration Key: `complaint_item_ref` / `item_ref`.

Core Entities:

`complaints`: Citizen complaint reports, adverse reactions, complaint severity (`critical`, `high`).

`ministry_blacklist`: Legally banned or seized counterfeit codes with immediate police seizure mandate.

`lab_reports`: Independent government chemical test results (`active_ingredient_match`, `contaminants_found`, result `pass` / `fail`).

`public_alerts`: Gazette safety advisories broadcast to consumers.

6. Multi-PC & Docker Distributed Deployment

In an enterprise environment, these databases reside on different machines across corporate firewalls.

Topology Configuration (`integration/distributed_config.py`)

`distributed_config.py` provides dynamic endpoint resolution:

```
# Reads environment variables with seamless fallback to localhost
PC1_HOST = os.environ.get("PC1_HOST", "127.0.0.1")
PC1_PORT = int(os.environ.get("PC1_PORT", "8011"))

PC2_HOST = os.environ.get("PC2_HOST", "127.0.0.1")
PC2_PORT = int(os.environ.get("PC2_PORT", "8012"))
DB4_PORT = int(os.environ.get("DB4_PORT", "8001"))
```

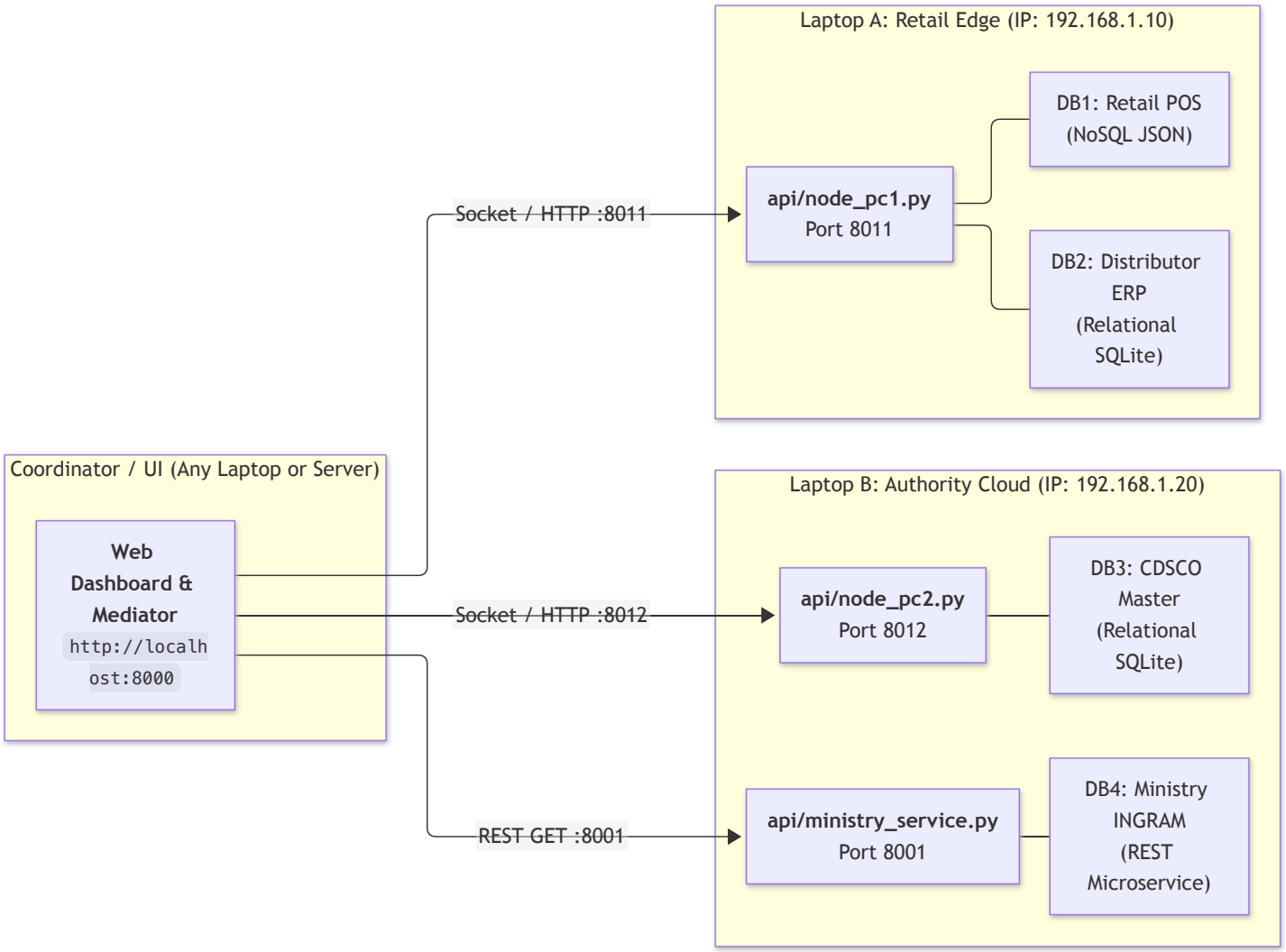
Docker Compose Architecture (`docker-compose.yml`)

```
services:
  asali-pc1-retail:
    build: .
    container_name: asali-pc1-retail
    command: ["python3", "api/node_pc1.py", "--port", "8011"]
    ports: ["8011:8011"]
    networks: [asali-network]

  asali-pc2-authority:
    build: .
    container_name: asali-pc2-authority
    command: ["python3", "api/node_pc2.py", "--port", "8012"]
    ports: ["8012:8012", "8001:8001"]
    networks: [asali-network]

  asali-mediator:
    build: .
    container_name: asali-mediator
    command: ["python3", "-m", "http.server", "8000", "--directory", "dashboard"]
    ports: ["8000:8000"]
    environment:
      - PC1_HOST=asali-pc1-retail
      - PC2_HOST=asali-pc2-authority
    networks: [asali-network]
```

To run on **two physical laptops on the same Wi-Fi**:



Laptop A (IP `192.168.1.10`): Run `python3 api/node_pc1.py --port 8011`.
Laptop B (IP `192.168.1.20`): Run `python3 api/node_pc2.py --port 8012` and `python3 api/ministry_service.py --port 8001`.
Laptop A or Server: Set `PC1_HOST=192.168.1.10 PC2_HOST=192.168.1.20 python3 -m http.server 8000 --directory dashboard`.

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7. The Modular Wrapper Layer

In Information Integration, a **Wrapper** encapsulates a specific data storage technology, translating high-level canonical query requests into native queries and returning standardized Python dictionaries.

All wrappers inherit from `BaseWrapper` (integration/wrappers/base_wrapper.py):

```
class BaseWrapper(ABC):
    def __init__(self, source_id: str):
        self.source_id = source_id

    @abstractmethod
    def fetch_by_key(self, key_value: str) -> Dict[str, Any]:
        """Fetch raw records matching the primary integration key."""
        pass

    @abstractmethod
    def health_check(self) -> bool:
        """Verify underlying connection / file availability."""
        pass
```

1. NoSQL Wrapper (`nosql_wrapper.py`)

Reads `data/nosql_store/db1_retail_pos.json`.

Streams documents into memory, evaluates predicates (`doc["scan_code"] == key`), and flattens nested structures.

Records execution latency in milliseconds for query telemetry.

2. Relational SQL Wrapper (`sql_wrapper.py`)

Used for DB2 and DB3.

Opens SQLite connection with `conn.row_factory = sqlite3.Row`.

Executes foreign key `JOIN` queries:

```
SELECT p.*, m.company_name, m.license_no, m.gmp_certified
FROM products p
JOIN manufacturers m ON p.manufacturer_id = m.manufacturer_id
WHERE p.product_uid = ?
```

3. REST API Wrapper (`rest_wrapper.py`)

Used for DB4 (Ministry Web Microservice).

Issues HTTP `GET` requests to `http://<host>:8001/api/v1/dossier?item_ref=<key>`.

Includes bearer authorization headers (`Authorization: Bearer CDSCO-MCA-2026-KEY`).

Implements a **Circuit Breaker** pattern: if the remote HTTP service drops, it gracefully logs network degradation and proxies to local storage without crashing the user session.

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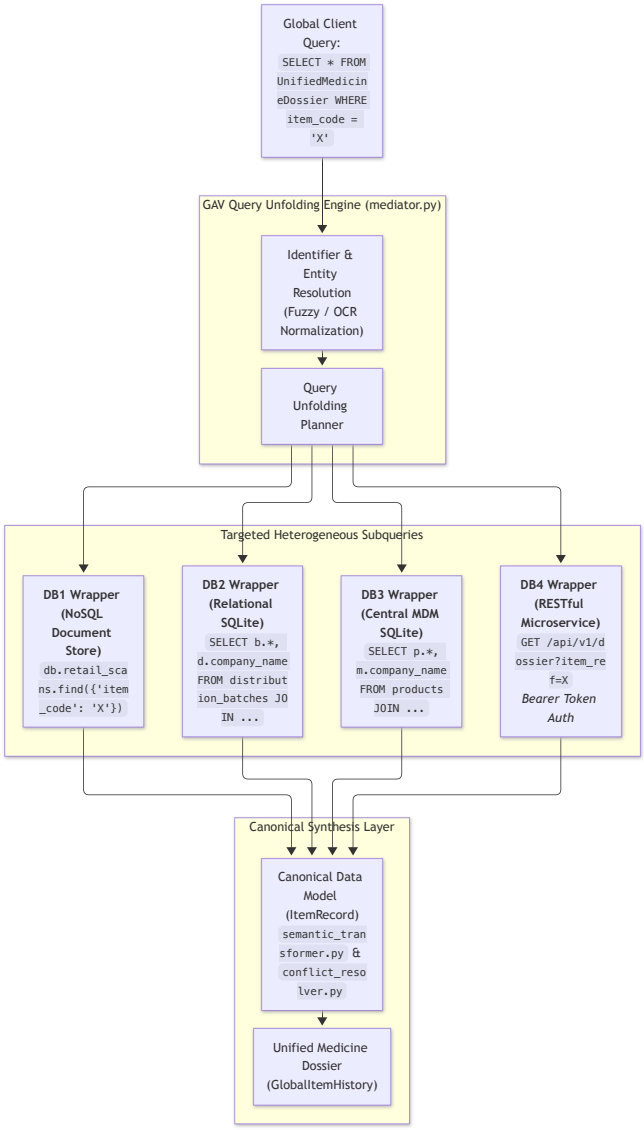
8. Global-as-View (GAV) Mediator & Query Unfolding

The core innovation of Asali is its **Global-as-View (GAV)** architecture implemented in `integration/mediator.py`.

In GAV, the **Global Schema** is defined as a view over the underlying **Source Schemas**:

$$\text{Global Schema } \mathcal{G} = \mathcal{V}(\mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3, \mathcal{S}_4)$$

When an application queries the global concept `UnifiedMedicineDossier`, the Mediator **unfolds** the query into physical subqueries:



Unfolding Implementation in Code (mediator.py):

```
def explain_query_unfolding(self, query_code: str) -> Dict[str, Any]:
    entity_res = self.entity_engine.resolve_item_code(query_code)
    canonical = entity_res.resolved_canonical

    return {
        "global_query": f"SELECT * FROM UnifiedMedicineDossier WHERE item_code =
'{query_code}'",
        "entity_linkage": {
            "raw_input": query_code,
            "canonical_target": canonical,
            "confidence_score": entity_res.confidence_score,
            "is_fuzzy_match": entity_res.is_fuzzy_linked
        },
        "subqueries": [
            {
                "target_source": "DB1 Retail Pharmacy POS",
                "engine": "NoSQL Document Store",
                "unfolded_operation": f"db.retail_scans.find({{'item_code': '{canon
ical}'}})",
                "purpose": "Detect duplicate scans, velocity anomalies, and unautho
rized vendor sales"
            },
            {
                "target_source": "DB2 Distributor ERP",
                "engine": "Relational SQL (SQLite)",
                "unfolded_operation": f"SELECT b.*, d.company_name FROM distributio
n_batches b JOIN distributors d ON b.distributor_id = d.distributor_id WHERE b.batc
h_serial = '{canonical}'",
                "purpose": "Audit supply chain custody, cold-chain temperature excu
rsion, transit route"
            },
            {
                "target_source": "DB3 CDSCO Master Registry",
                "engine": "Central MDM Catalog (SQLite)",
                "unfolded_operation": f"SELECT p.*, m.company_name FROM products p
JOIN manufacturers m ON p.manufacturer_id = m.manufacturer_id WHERE p.product_uid =
'{canonical}'",
                "purpose": "Verify statutory existence, authorized distributors, ba
tch certificate QC"
            },
            {
                "target_source": "DB4 Ministry INGRAM Portal",
                "engine": "HTTP RESTful Microservice",
                "unfolded_operation": f"GET /api/v1/dossier?item_ref={canonical}",
                "purpose": "Fetch active recalls, citizen grievance reports, labora
tory chemical tests"
            }
        ]
    }
```

9. Detection Logic: Two-Tier Hybrid Architecture

A critical innovation in Asali is resolving the **Subtractive Scoring Paradox**.

The Problem with Simple Additive/Subtractive Scoring:

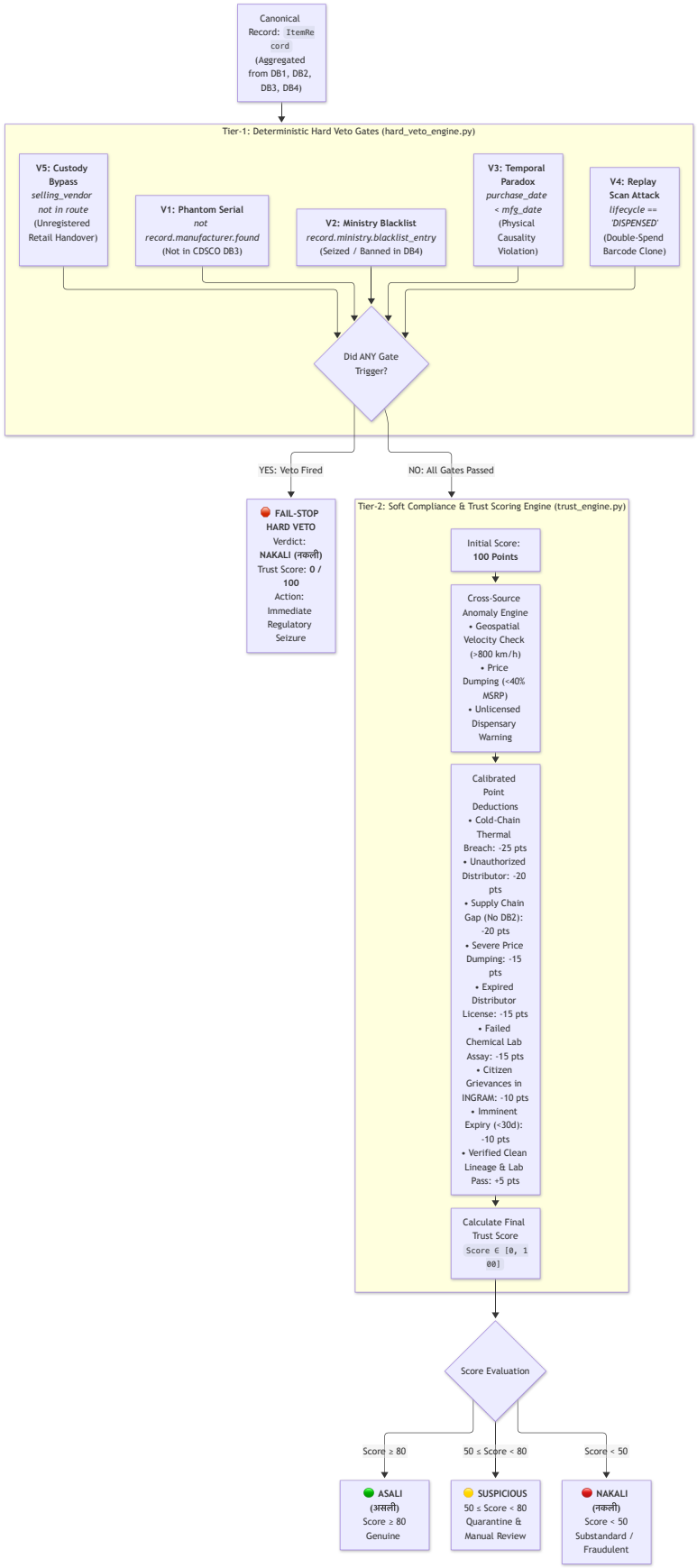
Imagine an engine that starts with a score of 100 and subtracts points for errors:

> If a medicine fails the manufacturer check: subtract 40 points.
> Score: $100 - 40 = 60$.
> Result: **60 points = "SUSPICIOUS - REVIEW REQUIRED"**.

This is unacceptable in healthcare. If a medicine was never manufactured by any licensed pharmaceutical company, it is not "suspicious" — it is a **100% fake (Nakali)**. It must have a score of 0 and trigger immediate police escalation.

The Solution: Two-Tier Hybrid Architecture

> **Tier-1: Hard Veto Gates** (Deterministic, zero-tolerance, fail-stop).
> **Tier-2: Soft Compliance Scoring** (Probabilistic, weighted deductions).



Tier-1: Deterministic Hard Veto Gates (`hard_veto_engine.py`)

If ANY of these 5 gates fire, evaluation halts immediately:

$$\text{Verdict} = \mathbf{NAKALI}, \quad \text{Trust Score} = 0$$

```
class HardVetoEngine:
    def evaluate(self, record: ItemRecord, lifecycle_state: Optional[str] = None) -
> VetoResult:
        gates = [
            self._v1_phantom_serial(record),
            self._v2_ministry_blacklist(record),
            self._v3_temporal_paradox(record),
            self._v4_replay_scan(record, lifecycle_state),
            self._v5_custody_bypass(record),
        ]
        triggered = [g for g in gates if g.triggered]
        return VetoResult(vetoed=len(triggered) > 0, gates=gates)
```

The 5 Hard Veto Gates:

- `V1_PHANTOM_SERIAL`:
  Logic: `not record.manufacturer.found`
  Meaning: The serial code exists in the retail shop (DB1), but was never registered in the Central CDSCO Manufacturer Database (DB3). It is a pure counterfeit manufactured in an illicit facility.
- `V2_MINISTRY_BLACKLIST`:
  Logic: `record.ministry.blacklist_entry and blacklist_entry["active"] == 1`
  Meaning: The drug code was explicitly seized, tested, and blacklisted by government gazette notice for lethal toxicity or illegal distribution.
- `V3_TEMPORAL_PARADOX`:
  Logic: `purchase_date < manufacture_date`
  Meaning: The point-of-sale customer receipt timestamp in DB1 is earlier than the factory manufacturing release date in DB3. This physical causality violation proves fraudulent barcode generation.
- `V4_REPLAY_SCAN` **(Double-Spend / Barcode Cloning)**:
  Logic: `lifecycle_state == "DISPENSED" and is_new_checkout_scan`
  Meaning: A 2D DataMatrix code on a prescription box can only be dispensed to a patient **once**. If that same code is scanned again at another counter, it is a cloned duplicate.
- `V5_CUSTODY_BYPASS`:
  Logic: `selling_vendor_id not in authorized_destination_vendors`
  Meaning: The medicine is being sold by a dispensary that was never part of the legal distributor dispatch route logged in DB2.

Tier-2: Anomaly Detection Engine (`anomaly_detector.py`)

When all Tier-1 gates pass, the **Anomaly Detector** inspects temporal and cross-source patterns:

```
class AnomalyDetector:
    def detect(self, record: ItemRecord) -> AnomalyReport:
        flags = []
        flags += self._check_multi_location(record)      # Barcode cloning across c
ities
        flags += self._check_temporal_consistency(record)# Handoff sequence
        flags += self._check_price_anomaly(record)       # Price dumping
        flags += self._check_velocity(record)            # Volume spikes
        flags += self._check_unlicensed_vendor(record)   # Rogue pharmacy
        flags += self._check_supply_chain_gap(record)    # Missing distributor
        flags += self._check_cold_chain_risk(record)     # Temperature breach
        return AnomalyReport(flags=flags, clean=len(flags) == 0)
```

Geospatial Velocity Feasibility Formula:

To detect barcode cloning, Asali calculates the minimum travel velocity between consecutive store scans:

$$v = \frac{\text{HaversineDistance}(\text{City}_1, \text{City}_2)}{\Delta t}$$

If $v > 800\ \text{km/h}$ (faster than commercial jet speed) between pharmacy scans, `MULTI_LOCATION_SCAN` fires automatically.

Tier-2: Probabilistic Soft Trust Scoring (`trust_engine.py`)

When Tier-1 passes, the drug begins with **100 Points** and is subjected to calibrated deductions:

```
score = 100

# 1. Cold-Chain Thermal Excursion (-25)
if requires_cold_chain and cold_chain_maintained == 0:
    score -= 25

# 2. Unauthorized Distributor Route (-20)
if not distributor.get("is_authorized", 1):
    score -= 20

# 3. Supply Chain Gap (-20)
# Item sold at retail but has zero distributor dispatch batch in DB2
if record.purchase.found and not record.distribution.found:
    score -= 20

# 4. Severe Price Dumping (-15)
# Sold at less than 40% of manufacturer MSRP
if price_paid < 0.40 * msrp:
    score -= 15

# 5. Revoked Distributor License (-15)
if distributor_license_expired:
    score -= 15

# 6. Failed Forensic Lab Test (-15)
if lab_report.get("result") == "fail":
    score -= 15

# 7. Citizen Complaints in INGRAM (-10)
if len(record.ministry.complaints) > 0:
    score -= 10

# 8. Imminent Expiry Date (-10)
if days_to_expiry < 30:
    score -= 10

# Bonus (+5)
if clean_lineage and lab_report.get("result") == "pass":
    score = min(100, score + 5)
```

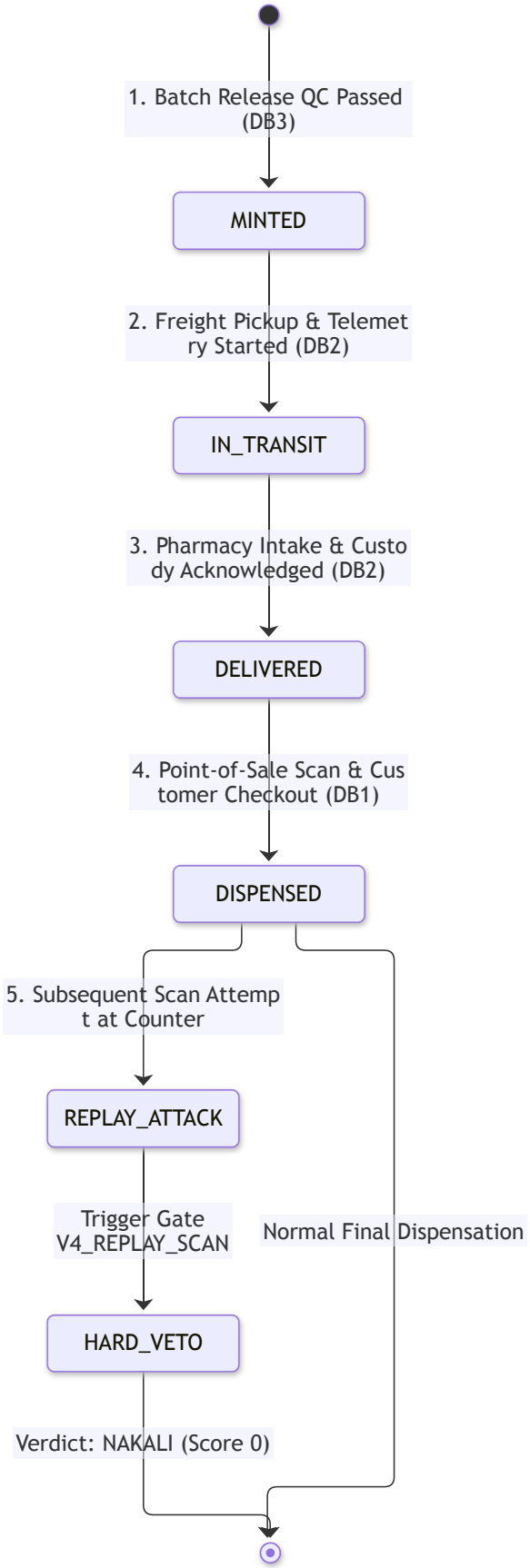
Verdict Decision Boundary:

$$\text{Verdict} = \begin{cases}
\mathbf{ASALI} & \text{if } \text{Score} \ge 80 \text{ and } \text{Tier-1 Passed} \\
\mathbf{SUSPICIOUS} & \text{if } 50 \le \text{Score} < 80 \text{ and } \text{Tier-1 Passed} \\
\mathbf{NAKALI} & \text{if } \text{Score} < 50 \text{ or } \text{Tier-1 Veto Fired}
\end{cases}$$

Lifecycle State Machine Tracking (`lifecycle_state.py`)

A legitimate medicine pack moves through a strictly sequential state machine:

$$\textbf{MINTED} \longrightarrow \textbf{IN_TRANSIT} \longrightarrow \textbf{DELIVERED} \longrightarrow \textbf{DISPENSED}$$



If an item's current state is `DISPENSED`, any subsequent checkout scan is flagged as a duplicate clone.

---

10. The 8 Information Integration & Architecture (IIA) Challenges

Asali is built as a complete academic research platform solving 8 core Information Integration challenges:

Challenge 01: Syntactic Heterogeneity & GAV Query Unfolding

Theoretical Problem: Disparate storage engines speak fundamentally different query languages: NoSQL JSON query predicates vs ANSI SQL 3NF JOINs vs RESTful HTTP GET.

Asali Implementation: mediator.py defines a Global-as-View query unfolding engine.

Code:

```
# Unfolds single canonical query into 4 distinct native syntaxes
unfolded_plans = {
    "db1": "db.retail_scans.find({'item_code': '...'})",
    "db2": "SELECT b.*, d.company_name FROM distribution_batches b JOIN distrib
utors d ...",
    "db3": "SELECT p.*, m.company_name FROM products p JOIN manufacturers m
...",
    "db4": "GET /api/v1/dossier?item_ref=... [Bearer CDSCO-MCA-2026-KEY]"
}
```

Challenge 02: Structural Heterogeneity & Hierarchy Flattening

Theoretical Problem: Schema mismatch between nested document trees (hierarchical 3D) and flat tabular rows (relational 2D).

Asali Implementation: nosql_wrapper.py and canonical_model.py.

Code:

```
# Safe path projection flattening nested NoSQL tree to 2D
flat_event = {
    "event_id": doc.get("_id"),
    "scan_code": doc.get("scan_code"),
    "vendor_id": doc.get("dispensary", {}).get("vendor_id"),
    "vendor_name": doc.get("dispensary", {}).get("name"),
    "city": doc.get("dispensary", {}).get("geo", {}).get("city"),
    "price_paid": doc.get("transaction", {}).get("amount"),
    "currency": doc.get("transaction", {}).get("currency", "INR")
}
```

Challenge 03: Semantic & Scientific Unit Heterogeneity

Theoretical Problem: Sensor telemetry uses different units of measure ($^\circ\text{F}$, K, $^\circ\text{C}$); currency is logged in foreign currencies (USD, EUR, KES, INR); date formats vary (Unix Epoch, Gazette `DD/MM/YYYY`, ISO-8601).

Asali Implementation: semantic_transformer.py.

Code:

```
@staticmethod
def normalize_temperature(temp_val: Any, source_unit: str = "C") -> Dict[str, A
ny]:
    val = float(temp_val)
    unit = source_unit.strip().upper()
    if unit in ("F", "FAHRENHEIT"):
        celsius = (val - 32.0) * 5.0 / 9.0
    elif unit in ("K", "KELVIN"):
        celsius = val - 273.15
    else:
        celsius = val
    return {"value_celsius": round(celsius, 2), "canonical_unit": "C"}
```

Challenge 04: Multi-Metric Fuzzy Entity Resolution & OCR Jitter

Theoretical Problem: Optical scanners swap glyphs (`O` ↔ `0`, `I` ↔ `1`); pharmacists make spelling typos (`"Novamax 500"` vs `"Novamox 500"`; `"Pasyamol"` vs `"Paracetamol"`).

Asali Implementation: entity_resolver.py and identifier_normalizer.py.

Code:

```
# 1. Optical Glyph Substitution
OCR_SUBSTITUTIONS = {'O': '0', 'I': '1', 'l': '1', 'Z': '2', 'S': '5', 'B':
'8'}

# 2. Multi-Metric Similarity Fusion (Fellegi-Sunter Paradigm)
sim = 0.50 * jaro_winkler(s1, s2) + 0.35 * levenshtein_ratio(s1, s2) + 0.15 * s
oundex_match(s1, s2)
if sim >= 0.72:
    return ResolutionMatch(canonical_code=catalog_key, confidence=sim)
```

Challenge 05: Declarative Conflict Arbitration & Contradictions

Theoretical Problem: Independent databases assert contradicting facts for the exact same item.

Asali Implementation: conflict_resolver.py enforces declarative precedence policies.

Code:

```
# Authority Precedence Policy: Government blacklist strictly overrules manufact
urer claim
if db4_blacklisted and not db3_recalled:
    conflicts.append(ConflictResolutionEntry(
        conflict_type="RECALL_AUTHORITY_CONTRADICTION",
        winning_source="DB4_MINISTRY_OF_CONSUMER_AFFAIRS",
        governing_policy="STATUTORY_DOMINANCE_POLICY",
        arbitrated_value={"is_recalled": True, "action": "IMMEDIATE_SEIZURE"}
    ))
```

Challenge 06: CAP Theorem, Network Partition & Circuit Breakers

Theoretical Problem: In a distributed deployment, physical cables disconnect or remote cloud containers crash. A naive mediator hangs or throws HTTP 500 errors.

Asali Implementation: Circuit breakers with dynamic penalty degradation in api/verify_item.py and rest_wrapper.py.

Code:

```
try:
    with urllib.request.urlopen(req, timeout=1.5) as resp:
        data = json.loads(resp.read())
except (urllib.error.URLError, TimeoutError):
    # Trip circuit breaker, fall back gracefully to local snapshot
    circuit_breaker.trip("PC1_NODE")
    # Penalize evidence completeness rather than crashing
    evidence_completeness -= 25  # Missing 1 of 4 database witnesses
```

Challenge 07: Schema Evolution & GS1 Digital Link Drift

Theoretical Problem: Remote nodes upgrade schemas asynchronously. When PC1 upgrades from Schema v1.0 (flat JSON) to Schema v2.0 (GS1 Digital Link URI `/01/gtin/21/serial`, minor-unit pricing, and GLN codes), static pipelines fail.

Asali Implementation: schema_adapter.py.

Code:

```
def adapt_retail_scan(doc: Dict[str, Any]) -> Dict[str, Any]:
    if "digital_link_uri" in doc:
        # Extract serial from GS1 Digital Link regex
        match = re.search(r"/21/([A-Z0-9\-]+)", doc["digital_link_uri"])
        doc["item_code"] = match.group(1)
        # Convert ISO-4217 minor currency units (4800 paise -> 48.00 INR)
        doc["price_paid"] = doc["transaction_pricing"]["amount_gross"] / 100.0
    return doc
```

Challenge 08: Transitive Multi-Database Identity Graph (Disjoint Silos)

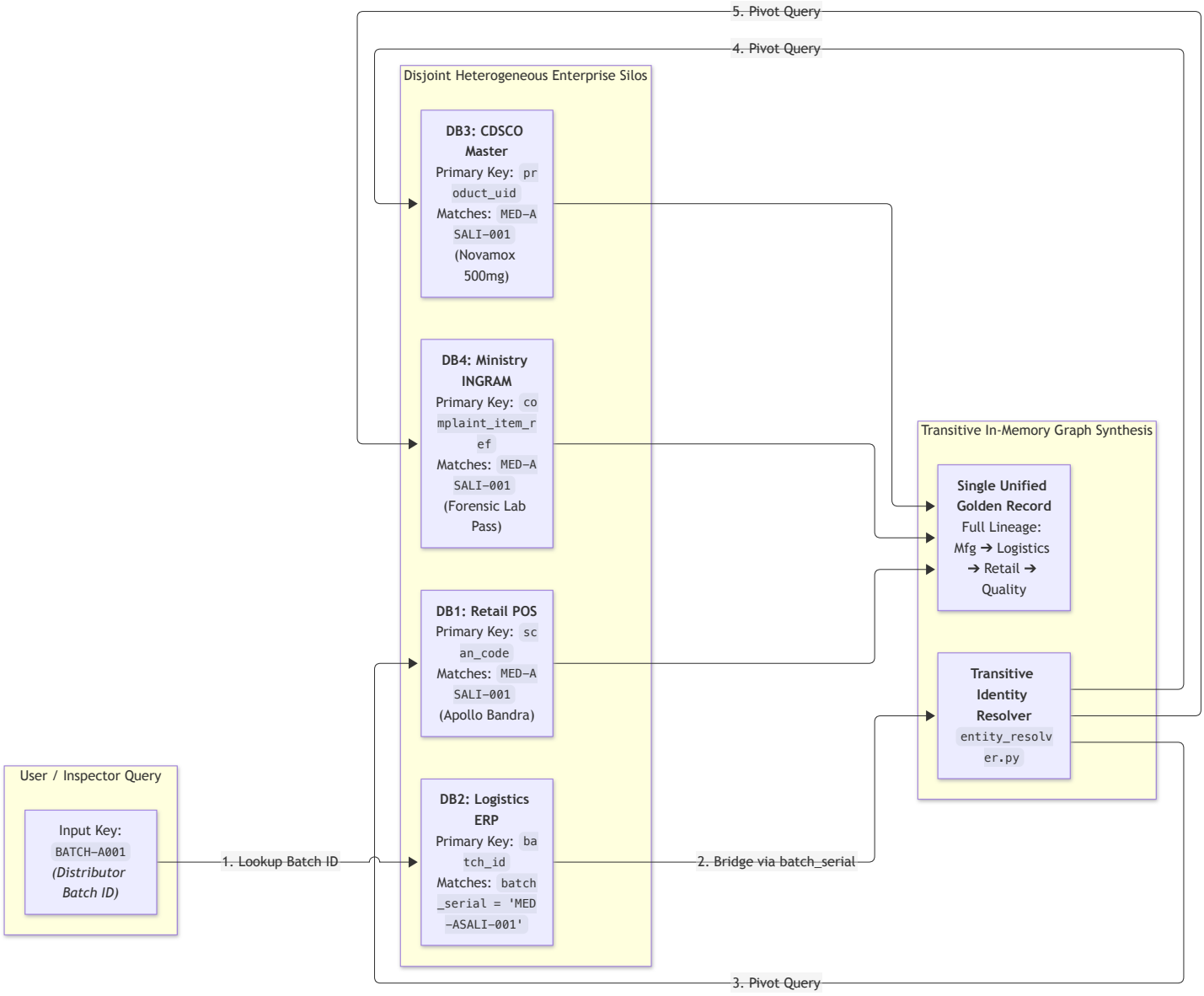
Theoretical Problem: Autonomous enterprise databases share **no universal primary key**.

DB1 (POS): Keys by `scan_code: "MED-ASALI-001"`.

DB2 (ERP): Keys by `batch_id: "BATCH-A001"` and `distributor_id`.

DB3 (CDSCO): Keys by `product_uid` and generic formula.

DB4 (Ministry): Keys by `complaint_item_ref`.



Asali Implementation: entity_resolver.py.

Code:

```
def resolve_transitive_identity(query_key: str) -> Dict[str, Any]:
    # Step 1: Query DB2 by Batch ID to find associated serial numbers
    batch_records = query_db2_by_batch(query_key)
    canonical_serial = batch_records[0]["batch_serial"]

    # Step 2: Traverse in-memory transitive graph
    # BATCH_ID -> SERIAL_CODE -> PRODUCT_UID -> DOSSIER
    return {
        "canonical_id": canonical_serial,
        "linkage_path": ["DB2_BATCH_ID", "DB1_SERIAL_CODE", "DB3_PRODUCT_UID"],
        "golden_record": synthesize_golden_record(canonical_serial)
    }
```

11. Cryptographic Immutable Audit Chain

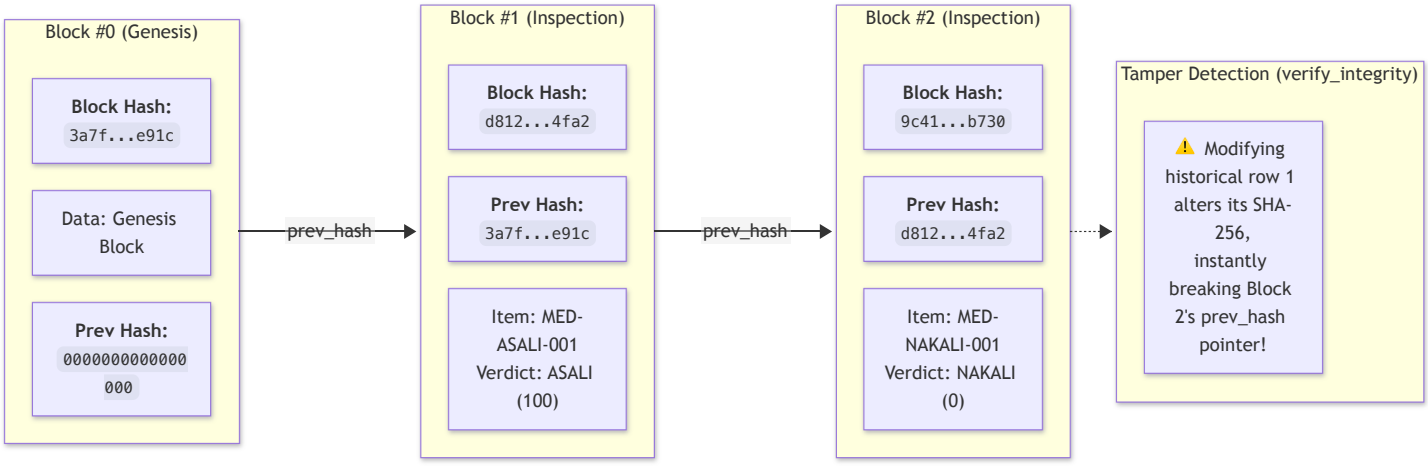
To guarantee that corrupt actors cannot alter past verification logs, Asali includes a built-in cryptographic audit ledger in integration/audit_chain.py.

Block Structure (`databases/audit_chain.sqlite`):

```
CREATE TABLE IF NOT EXISTS audit_blocks (
    block_index   INTEGER PRIMARY KEY AUTOINCREMENT,
    timestamp     TEXT NOT NULL,
    item_code     TEXT NOT NULL,
    verdict       TEXT NOT NULL,
    trust_score   INTEGER NOT NULL,
    db_sources    TEXT NOT NULL,
    prev_hash     TEXT NOT NULL,
    block_hash    TEXT NOT NULL UNIQUE
);
```

Cryptographic Chaining Formula:

$$\text{block_hash} = \text{SHA-256}\Big(\text{index} \,\|\, \text{timestamp} \,\|\, \text{item_code} \,\|\, \text{verdict} \,\|\, \text{trust_score} \,\|\, \text{prev_hash}\Big)$$



If a single row is modified or deleted by an attacker, `verify_chain_integrity()` detects the broken hash pointer and alerts administrators.

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12. Frontend Architecture & Tactile Ceramic UI System

The user interface is built strictly in **Vanilla HTML5, Vanilla JavaScript, and Tailwind CSS**, adhering to high-end design aesthetics:

- **Zero Component Borders**:
  Strict removal of artificial hairline borders (`border border-black/[0.04]`), preventing cluttered wireframe aesthetics.
- **Small Minimal Tactile Shadows**:
  Custom layered drop shadows:

  ```
  box-shadow: 0 2px 8px -1px rgba(0, 0, 0, 0.07), 0 1px 3px rgba(0, 0, 0, 0.04);
  ```

  Grounding Layer (`0 1px 3px`): Provides razor-sharp edge definition separating cards from white/gray surfaces.

  Ambient Lift (`0 2px 8px`): Subtle downward elevation with negative spread choke to prevent blurry halos.

  Hover Lift (`hover:shadow-[0_6px_16px_-2px_rgba(0,0,0,0.1),0_2px_5px_rgba(0,0,0,0.04)]`): Interactive elevation.
- **Octane / Blender 3D Ceramic Art**:
  Dedicated 3D dioramas for all 8 challenges (e.g. `challenge4_fuzzy_3d.png` , `challenge8_nanobanana_3d.png`).
- **Interactive Verification Pipeline**:
  4-step status progress indicator (`Barcode` → `Cold-Chain` → `Manufacturer` → `Ministry`).

  Radial SVG Trust Gauge (0 to 100).

  Live distributed topology console displaying real-time latencies across Node PC1 and Node PC2.

---

13. Step-by-Step Execution, Verification & Viva Cheat-Sheet

1. Initialize the Environment

```
# Step 1: Create all 4 SQLite database schemas
python3 create_databases.py

# Step 2: Seed relational databases
python3 data/seed_db3.py  # Manufacturer catalog
python3 data/seed_db2.py  # Distributor logistics
python3 data/seed_db1.py  # Retail POS scans
python3 data/seed_db4.py  # Ministry grievance reports

# Step 3: Seed NoSQL document store
python3 data/seed_nosql_db1.py
```

2. Run the Web Dashboard

```
python3 -m http.server 8000 --directory dashboard
```

Open: http://localhost:8000/index.html or http://localhost:8000/verify.html.

3. Run Distributed Multi-PC Cluster

```
python3 scripts/run_cluster.py
```

4. CLI Verification Commands

```
# Verify a genuine medicine
python3 api/verify_item.py MED-ASALI-001

# Verify a counterfeit medicine (triggers Tier-1 Hard Veto)
python3 api/verify_item.py MED-NAKALI-001

# Verify a recalled batch
python3 api/verify_item.py MED-RECALL-001

# Verify with automated HTML report generation
python3 api/verify_item.py --report MED-ASALI-001

# Verify cryptographic audit chain integrity
python3 api/verify_item.py --audit
```

The Top 5 Questions You Will Be Asked in a Viva / Exam

Q1: What is the difference between GAV (Global-as-View) and LAV (Local-as-View)? Why did you pick GAV?

> **Answer**: In **GAV**, the global schema is defined as a view or query over the underlying source tables ($G = V(S)$). Query unfolding is straightforward because the mediator knows exactly which subquery to send to which source. In **LAV**, each local source is defined as a view over the global schema ($S = V(G)$), which requires complex query rewriting algorithms (like the Bucket Algorithm). Since pharmaceutical registries have stable global concepts (manufacturer, batch, scan, alert), GAV provides deterministic, low-latency unfolding.

Q2: What is the Subtractive Scoring Paradox and how did you solve it?

> **Answer**: In additive/subtractive systems, a drug starts at 100 points and loses points per failure. A completely fake drug that was never registered by any manufacturer might only lose 40 points, scoring 60 points ("Suspicious"). This is unacceptable in healthcare. We solved it with a **Two-Tier Hybrid Architecture**: Tier-1 executes 5 deterministic Hard Veto Gates (`V1_PHANTOM_SERIAL`, `V2_MINISTRY_BLACKLIST`, etc.). If any gate triggers, the score is forced to 0 and verdict to `NAKALI` immediately, completely bypassing soft scoring.

Q3: How do you link records when databases share no universal primary key? (The Airtel / Disjoint Silos Case)

> **Answer**: Real-world silos (POS vs ERP vs CDSCO) do not share keys. Asali builds an in-memory **Transitive Multi-Database Identity Graph** in `entity_resolver.py`. It uses secondary keys to bridge silos: querying DB2 by `batch_id` yields `batch_serial`, which bridges to DB1's `scan_code` and DB3's `product_uid`. Combined with Fellegi-Sunter probabilistic matching (0.72 threshold), it synthesizes a single unified Golden Record.

Q4: How does your system handle network partition (CAP Theorem)?

> **Answer**: When a physical LAN cable disconnects or a remote node drops, a naive mediator hangs or throws a 500 error. Asali implements a **Circuit Breaker** pattern in `rest_wrapper.py` and `verify_item.py`. When a node fails to respond within $1.5\ \text{seconds}$, the mediator marks the node `OFFLINE`, falls back to local cached snapshots, and transparently penalizes the `evidence_completeness` percentage rather than crashing the user session.

Q5: How do you prove that past audit records were not altered by a corrupt employee?

Answer: Every verification event is appended to `databases/audit_chain.sqlite` as a cryptographic block. Each block contains a SHA-256 hash calculated over the current inspection details concatenated with the `prev_hash` of the previous block. Modifying any historical row breaks the hash chain, which is instantly detected by `audit_chain.py:verify_integrity()`.

Document compiled and verified for the Asali / Nakali Information Integration & Architecture (IIA) Project.

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